25901145 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BBA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBBA200 - STATISTICS FOR BUSINESS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

1. Define Statistics. [U] / [CO1]
2. Mention one limitation of the semi-average method. [K] / [CO2]
3. What is cyclic variations? [K] / [CO2]
4. What are the merits and demerits of least square method? [K] / [CO2]
5. Explain the difference between a paired t-test and an independent samples t-test. [U] / [CO3]
6. Define type I error. [K] / [CO3]

[3x5 = 15]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

7. The following table gives the time series data(quarterly sales) and the estmated trend values (T): [K] / [CO2]

Quarter	Sales (Rs.000)	Trend(Rs.000)
Q1	110	120
Q2	150	140
Q3	190	160
Q4	130	180

Find the seasonal component (S) for each quarter assuming an additive model (ignore cyclical and irregular variations).

8. What is meant by a 4 year and 5 year moving averages? [A] / [CO2]

9. What are the applications of t-tests? [A] / [CO3]

10. Write the working procedure for Kruskal-Wallis test. [A] / [CO3]

[5x3 = 15]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

11. ' Statistics is a tool for decision making' - Explain. [U] / [CO1]

12. The following table shows yearly profit [A] / [CO2]

Year	2014	2015	2016	2017	2018	2019	2020
Profit	40	45	50	48	55	60	65

Fit a trend line by free hand curve method.

13. The number of misprints found in books printed by three printing presses are recorded below. Use One-way ANOVA to test whether there is a significant difference. [A] / [CO3]

Press A	Press B	Press C
4	5	6
5	6	7
6	7	8

14. A survey of radio listeners' preference for two types of music under various age groups gave the following information. Is preference for type of music influenced by age? Use 5% level of significance. [A] / [CO3]

Type of Music	19–25	26–35	Above 36
Carnatic music	80	60	90
Film music	210	325	44
Indifferent	16	45	132

[10x2 = 20]